MILITARY SPECIFICATION

STAND, TEST, AIRCRAFT HYDRAULIC SYSTEMS, NON-PORTABLE

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for test stands for aircraft hydraulic systems and components and shall become a part of a detail specification by attachment or reference.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifictions and standards of the issue listed in that issue of the Department of Defense index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

CC-M-1807	Motor Alternating Current, Fractional and Integral Horsepower (500 HP and Smaller)

MILITARY

MIL-C-104	Crate, Wood, Lumber and Plywood Sheathed, Nailed and Bolted
MIL-P-116	Preservation, Methods of
MIL-A-5498	Accumulators , Aircraft Hydropneumatic Pressure
MIL-H-5606	Hydraulic Fluid, Petroleum Base, Aircraft, Missile, and Ordnance
MIL-P-6808	Primer Coating, Zinc Chromate, Process For Application of
MIL-W-10434	Window Observation

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to the Engineering Division, San Antonio Air Logistics Center, Kelly AFB, Texas, 78241 by using the self-addressed Standardization Document ‘Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4920

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:09Z
Check the source to verify that this is the current version before use.

STANDARDS

FEDERAL

	FED-STD-595	Colors

MILITARY

	MIL-STD-129	Marking for Shipment and Storage
	MIL-STD-130	Identification Markings of U.S. Military Property
*	MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
*	MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
*	MIL-STD-808	Finishes. Protective, and Codes, for Finishing Schemes for Ground and Ground Support Equipment
*	MIL-STD-889	Dissimilar Metals
*	MIL-STD-1186	Cushioning Anchoring, Bracing, Blocking and Waterproofing , with Appropriate Test Methods

Air Force-Navy Aeronautical

AN6287	Valve , Air High Pressure 3/4 Hex
AND20001	Drive, Type XI Engine Accessory
AND20002	Drive , Type XII Engine Accessory

* (Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.1.2 Order of precedence. In the event of a conflict between the text of this specification and the reference cited herein, the text of this specification shall take precedence.

* 2.2 Other publications . The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

*NATIONAL FIRE protection ASSOCIATION

NFPA 70	National Electrical Code (Current Issue)

*(Copies of National Fire Protection Association publications may be obtained from the National Fire Protection Association, 60 Batterymarch St., Boston MA 02110.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for the preproduction testing.

3.2 Components. The stand shall consist of the following major components : hydraulic system, electrical system, pneumatic system, cooling system, and all accessories thereto necessary for proper and efficient operation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:09Z
Check the source to verify that this is the current version before use.

3.3 Selection of specifications and standards. Specification and standards for necessary commodities and services not specified herein shall be selected according to MIL-STD-143 except as provided in 3.3.1.

3.3,1 Standard parts, AN or MS standard parts shall be used wherever they are suitable for the purpose and shall be identified by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, hose, fittings, etc., may be used, provided they have suitable properties and are Interchangeable with AN, MS, or MIL standard parts without alterations,

3.4 Material. All material used shall be as specified herein. Material not specified shall be of the best quality used for the purpose in good commercial practice. The material shall be free from all defects and imperfections that might affect the serviceability of the finished product. Mounting pads, adapters, and spline drives shall be constructed of material specified in the detail specification.

3.4.1 Protective treatment. Materials used in the construction of the equipment that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification.

3.4,2 Metals, Metals shall be of corrosion resistant type, unless suitably protected to resist corrosion during normal service life and shall be suitable for the application intended, Metals not resistant to corrosion shall be cadmium, dull zinc, or dull nickel plated.

3.4.2.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other, Dissimilar metals are defined in MIL-STD-889.

3.4,3 Plating. All fixtures, adapters and couplings that are not made of corrosion resistant materials or anodized aluminum shall be cadmium, dull zinc, or dull nickel plated, Plating shall be in accordance with standard commercial practice for this type of equipment.

3.5 Design and construction

3.5.1 Design. The design, layout and assembly of the cabinet and its component parts shall be such as to facilitate quantity production and ease of maintenance.

3.5.2 Structure All equipment shall be located within an all-metal fully closed cabinet suitably braced to carry the load imposed, The cabinet shall be fully guarded to shield fan blades, heating coils, electrical connections , and any other component parts that may present a hazard to the operator . The cabinet shall be so constructed that it will withstand transportation by common carrier without damage. Instrument panels shall be so constructed of 11 gage or heavier metal suitable for the purpose, Doors and cabinet panels other than instrument panels shall be a minimum of 14 gage metal.

3,5.3 Size. Cabinet size shall not exceed 10 feet length by S feet depth (depth from operator’s side of stand to rear) by 6 1/2 feet height. Test equipment requiring a cabinet of a larger size shall be constructed in sections to meet this size requirement,

Source: https://assist.dla.mil -- Downloaded: 202640-20T10:09Z
Check the source to verify that this is the current version before use.

3 .5.4 sections. Equipment that must be constructed in sections to meet the size requirements shall be so designed that disassembly for shipment and subsequent reassembly shall not create an alignment problem of the components affected All mating parts hydraulic and pneumatic lines, and electrical connections shall be properly identified by permanent type tags on nameplates to insure and facilitate proper reassembly.

3.5.5 Equipment mounting. Equipment such as instruments or ventilating fans, may be mounted outside the cabinet, increasing the overall dimensions , if readily removable for transit purposes.

* 3.5.6 Base design All cabinets and sections shall include longitudinal channels or other approved base design to permit the use of a standard fork-lift truck for positioning or moving purposes. These provisions shall be a permanent part of the unit of sufficient strength and rigidity to support the load without undue strain or stress on the equipment itself, and so arranged as to distribute the floor load evenly over the floor space occupied by the test unit, Channels (or other design as applicable) shall be a minimum of 8 inches width and spaced a maximum of 20 inches between centers. Two forklift openings shall be provided on the two opposite sides of each test stand or section The centerline of each opening shall be a maximum of 27+1/2 inches from, the center of gravity of the test stand or section and shall be equally spaced, Each opening shall be a minimum of 2.5 inches high 9.75 inches wide and capable of taking fork lengths up to 80 inches long.

3.5.7 Accessibility. The cabinet shall be of sufficient size to accommodate all equipment in an accessible manner. All internal components of the test equipment shall be accessible through removable hinged doors at the rear and sides with quick detachable hinges and provided with positive flush type or streamline hand latches which are easily operated.

3.5.8 Hinge pins , The hinge pins where used shall be tapered (by grinding) for ease of installation and the center hinge pin (or pins) shall be I/L-inch shorter than the top and bottom pins The top hinge pin shall be of the retractable spring loaded type, and the bottom pin the fixed type.

3.5.9 Opening, Doors shall not exceed 18 incl)es width, unless hinged in the center so that the door when open will not extend more than 18 inches from the sides or ends of the unit. Openings shall be of adequate size and properly located to allow ease of removal of any component in the stand. Doors shall be so hinged as to be removed or installed readily by one man.

3.5.10 Toe clearance, Toe clearance shall be provided at the operator’s position on the test unit, a minimum of 6 inches height and 4 inches depth,.

3.5.11 Storage space. A suitable storage container shall be provided for the operating manual and handbooks covering the test unit. This container shall be permanently attached to the test stand and in a readily accessible location.

3.5.12 Leveling. Provisions for lagging and leveling shall be as outlined in the detail specification

3.5.13 Working area Flat table top working areas shall be provided at each end of the sump drain, a minimum of 18 inches length by 18 inches width and shall be a minimum of 30 inches and a maximum of 34 inches above the floor,

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:09Z
Check the source to verify that this is the current version before use.

Test units constructed in more than one section shall have individual working areas integral with each section. The working area and sump drain combination shall be so designed and located as to facilitate installation, removal and visual inspection of accessories undergoing test

3.5.14 Vibration. The cabinet shall be constructed that vibration from mechanisms operating the test stand shall not be transmitted through the structural members to the instrumentation, thus imparing readability and accuracy,

* 3.5,15 Lights. If found to be required, lights shall be vapor proof, of ample wattage, and unless otherwise specified, be powered by 110 volt, single phase, 60 HZ current. Guards shall be provided around light bulbs to prevent personnel contact and inadvertent bulb breakage.

3.5.16 Integral lighting, Instrumentation shall be selected insofar as practicable to eliminate need for integral lighting.

* 3.5.17 Ventilation. Louver type air vents shall be provided in the cabinet near floor level and in the top to promote air circulation, or the cabinet shall be provided with forced ventilation if found to be required for any reason such as prevention of overheating of instruments and other electrical/electronic components.

3.5,18 Drip pan. A drip pan or pans shall be provided within the inclosure . The pan shall be installed to catch all leakage which may drip from any part or flow down the inside walls of the inclosure. Drip pan(s) shall be constructed of not less than 14 gage metal. Pan(s) shall have a minimum of 1 inch slope toward the center of the unit (or section). (The center is to be interpreted as the center of the instrument panel section(s) normally facing the operator.) A channel (or gutter) shall be incorporated into the center of each pan which shall in turn have a minimum of 2 inches slope from the front to the rear of the stand, Channel(s) (or gutters) shall terminate with not less than 1 inch female pipe coupling(s) at the rear of the test equipment unit , a minimum of 2 inches above the floor, Drain(s) shall be installed in an accessible location for cleaning, Pans shall be adequately reinforced to location for cleaning. Pans shall be adequately reinforced to withstand, without permanent distortion, the combined weight of repairman and components being removed and installed,

3.5.18.1 Equipment sections. Units constructed in sections shall have individual drip pans installed in such a manner to prevent leakage between sections and drip pans shall be installed and supported in such a manner to insure complete drainage of each section.

3.5.19 Test fixtures. Test fixtures shall be constructed in such a manner as not to interfere with adjustments that are required during calibration. Test fixtures for test stands designed to test only one type of accesory shall be furnished as component parts of the stand and identified as additional details to the basic test stand drawings.

3.5,19.1 General purpose. Various test fixtures for test stands designed to test a multiple number and type of accessories shall not be included with the basic test stand, with the exception of the base mounting fixture which shall be furnished as a detail to the basic test unit. Additional fixtures shall be furnished as test sets under separate manufacturers' part numbers.

Source: https://assist.dla.mil -- Downloaded: 202640-20T10:09Z
Check the source to verify that this is the current version before use.

3.5.20 Panel holes, All holes through vertical panels for installation of valves and fittings (and through horizontal surfaces normally described as the working table or area) shall be of a standard size and type to permit interchange of valves and fittings from 1/8 inch to 1 inch size, without rework of panel. Interchangeable flanged adapters shall be fitted to the panel holes and shall provide a rigid mounting of sufficient strength to hold the valve or fitting without addit ional support. Bevel plates or other means shall be provided for each panel mounted valve or fitting which will seal the opening as effectively as practicable, Plates shall be readily removable by one man.

3.5.21 Drive pad. All drive pads to which units will be attached for test shall be designed so that the mounting pad and spline adapter w1ll be fully removable and interchangeable with other sizes of splines and pads. The pads and splines shall be easily removed and installed from the front of the stand by one man. Pads and spl ines shall be in accordance with Standards AND20001 and AND20002, as appl icable

3 .5.21.1 Location. Mount ing pads and brackets shall be conveniently located in relation to the related instruments and controls.

3 .5.21.2 Materials. Mount ing pads , adapters and sp line drives shall be constructed of material specified in the detail specification,

3 .5.21.3 Quick change clamps. All drive pads shall be equipped with quick change clamps and unless otherwise specified shall have a dowel pin for each hole in the accessory mounting flange. Quantity of quick change clamps required, design, and configuration shall be as outlined in the detail specification.

3 .5.21.4 Dowel pins. Diameter, length of dowels , and material used shall be as specified in the detail specification.

3.6 Performance. The test stand shall be capable of operation through the specif ied capacity, temperature range, and pressure range for the period of time and under the actual test condit ions as called out under the requirements of the detail specification.

3.7 Detail of components

* 3. 7.1 Electrical system. The test stand electrical system shall be designed to supply power to the test stand components and the item under test, as required. Unless otherwise speicf ied in the detail specification, the electrical system shall be designed to operate from a 230/460 volt, 3 phase, four wire, 60 HZ electrical power source. The electrical system shall consist of all electrically operated components within the test stand.

* 3 .7.1.1 Motors. All motors shall be drip Proof and shall conform to CC-M-1907 . The motors shall be of adequate rating to operate under any condi- tions specified herein at maximum capaci. ty without exceeding the NEMA recom- mended temperature limits for the class of insulating material used, The motor manufacturer’s recommended full-load current shall not be exceeded during operation.

Source: https://assist.dla.mil -- Downloaded: 202610-20T10:09Z
Check the source to verify that this is the current version before use.

* 3.7.1.2 AC power. Unless otherwise specified, AC motors shall operate on 230/460 volt, 3 phase, 60 HZ current and shall be connected for operation on 460 volt power . Change over from 460 to 230 volts shall be possible with the minimum amount of addit ional parts. The electrical wire used shall be capable of both 230 and 460 volt operation. Unless otherwise specified, AC motors of 1/3 horsepower or less may be operated on 110-volt single phase, 60 HZ current by use of a transformer (see 3.7.1.3), with the primary leads arranged in such a manner that they can be selectively connected to any of the three phase legs of the power supply to allow for load balance when the unit is installed in conjunction with other components.

* 3.7. 1.2.1 AC control. The AC controller shall include an AC across the line starter for the main power unit for motors less than 25 horsepower. Motors 25 horsepower and above shall be provided with reduced voltage starters. Starters shall be capable of withstanding openings under starting surges. The starters shall include overload relays for motor protection that can be manually reset without opening the starter case.

* 3.7.1.3 Transformer. A transformer to supply the 110 volt power for lighting and control circuits shall be provided as a component of the test stand. The current rating for continuous operation of the transformer shall be sufficient to adequately power the 110-volt circuits of the test stand without overloading, and the operating temperature of the transformer shall be within the range recommended by the transformer manufacturer.

* 3.7.1.4 Push button stations. Unless otherwise specified, push button stations shall operate on ll0-volt, single phase, 60 HZ current,

3.7.1.5 Interlocks. Where there is more than one motor in the equipment and possible damage may be caused by the stoppage of any one motor, interlocks shall be provided so that the stoppage of one motor in the circuit will not endanger or damage the equipment. Where the starting of the motors in an improper sequence can result in damage, interlocks shall be provided to insure starting of all motors in proper sequence. Where cooling water or lubrication are required for components of the test unit such as vacuum pump or gear cases, the controls for such shall be automatic or shall be interlocked with the motor controls in a manner to prevent damage to the equipment through operating error or an over temperature or over pressure condition.

3.7.1.6 Wiring. All wiring conduit, switches, and other standard electrical components of the test unit shall conform to the applicable requirements of the (current edition) National Electrical Code.

3.7.1.7 Electrical connections. All electrical power sources connections to the equipment shall be made to terminal strips in a junction box located so that wires may enter from the top of the test equipment unit. The wires shall be coded to conform with the code numbers of the wiring diagram. An additional junction box shall be provided for lighting and control circuits.

Source: https://assist.dla.mil -- Downloaded: 2026l0-20T10:09Z
Check the source to verify that this is the currentversion before use.

3.7.1.8 DC circuit Where required to operate solenoids or other electrical components a 30-volt DC circuit shall be furnished with variable rheostat ammeter, voltmeter, pilot light, switches and other necessary components as provided in detail specifications.

* 3.7 .1.8.1 DC Components. The 30 volt DC supply from the stand shall be controlled by a momentary contact spring return type switch which cannot be energized unless continuously depressed manually. All cables supplying voltage to the item being tested shall terminate with a female connection.

* 3.7 .1,8.2 DC Power Supply. The 30-volt DC power supply shall be provided as a component of the test equipment. The power supply shall include means for maintaining the set voltage constant to within plus or minus 1/2 volt.

3.7.1.9 400 Cycle AC circuit Where required, a 400-cycle AC circuit shall be provided which shall include all necessary instrumentation such as wiring, receptacles, controls, ammeters, voltmeters, etc. Type, range and capacity of Instrumentation shall be as outlined in the detail specification

3. 7.1.9.1 400 Cycle AC Power Supply. The 400-cycle AC power supply shall be provided as a component of the test equipment. Type and capacity of the power supply shall be governed by the detail specification,

* 3.7.1.10 Emergency switch. Unless otherwise specified, two emergency shut off switches shall be provided on all test equipment units exceeding 8 feet in length. Each emergency switch shall be of the red, palm operated, mushroom - head, push button type and shall de–energize all electrical, air, water and hydraulic systems simultaneously. One switch shall be located near each end of the stand in a readily accessible position. On the shorter test stands a single switch shall be located at the operators normal position.

3.7.1.11 Pilot lights. Green pilot lights shall be provided on the basic test unit to indicate when each electrical circuit is turned on and operating properly .

3.7.1.12 Warning lights. A red light (in addition to a warning placard) shall be provided to indicate a dangerous condition, such as circuits energized to electrical accessories under test which may inadvertently be broken or disconnected by the operator.

3,7.1.13 Installation. All electrical equipment shall be installed in accordance with the applicable requirement of the National Electrical Code.

3.7.1.14 Grounding, All electrical equipment in the test stand shall be grounded in accordance with Article 250 of the National Electrical Code. A plainly marked grounding connection shall be provided for connection to the ground system.

Source: https://assist.dla.mil -- Downloaded: 2026t0-20T10:09Z
Check the source to verify that this is the currentversion before use.

* 3.7.1.15 Belt drives. Belt driven equipment shall not be used.

3.7.1.16 Vibration All motor sets shall be shock mounted to reduce vibration and noise to minimum. The shock mount ing shall effectively insulate the assembly from the test unit so as not to transmit any perceptible vibration to the installed instruments.

* 3.7.1.17 Variable speed drives. Variable speed drives shall be solid state electronics controlled DC motor drives and shall be suitable for the intended application, Minimum and maximum operating speeds, horsepower values for the various operating speeds, type of drive (reversing or nonreversing). and type of speed regulation (non–automatic or automatic), shall be as specified in the detail specification.

3.7.1.18 Controls. The speed control shall provide infinitely variable adjustment throughout the speed range. Allowable hysteresis, creep or backlash shall be as specified in the detail specification and shall be one of the classes listed in 3.7.1.18.1 below.

3.7 .1.18.1 Regulation. Automatic speed regulation shall be specified in the specification and shall be one of the following classes:

Class 1	0.5 percent
Class 2	0.25 percent
Class 3	0.1 percent
Class 4	0.04 percent

* Speed Regulation = $\frac{\text{(No load RPM) - (Full load RPM) x 100}}{\text{(No load RPM)}}$

* 3.7.1. 18.2 Response. The maximum deviation from the set speed, upon sudden variation of load shall not exceed 1 percent of the set speed. This deviation shall occur in the direction of load change only, with no subsequent overshoot. All transients resulting from sudden load variations or sudden changes in the set speed shall be effectively damped out in less than 0.5 second.

* 3.7 .1.18.3 DC control. The DC control shall be suitable for operation over the specified speed and horsepower range.

* 3.7 .1.18.4 Field loss relay. A motor field loss relay shall be provided, which shall shut the drive down should the motor lose its field. Provision shall be included to prevent the drive motor from being started, unless the speed setting rheostat is set for minimum speed,

3.7.2 Hydraulic System

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:09Z
Check the source to verify that this is the current version before use.

3.7.2,1 Reservoirs. All reservoirs shall be of corrosion resistant steel or aluminum alloy suitable for the application and provided with a clean-out plate readily accessible for removal. Reservoirs shall be of welded construction and provided with the necessary inlets, outlets, vents and gage connections. Stainless steel, brass or non-corrosive fittings shall be attached by atomic-hydrogen welding brazing, silver soldering, or other suitable maethods, as approved by the procuring agency. Method of attaching fittings shall be accomplished in a manner to reduce the possibility of electrolytic corrosion or erosion to a minimum. All reservoirs shall be thoroughly neutralized and cleaned for removal of weld flux, scale, and other foreign matter from the interior and exterior of the reservoir. Reservoir shall be sloped to permit complete drainage of the fluid by gravity. A sump shall be provided below the inlet port to the pump.

3.7.2. 1.2 Baffles. Baffles, if required, shall be installed in such a manner as to prevent pressure pulsations which would affect accuracy and shall be so designed to reduce turbulence and aeration to a minimum. Baffles shall be designed and located with sufficient clearance at the bottom of the reservoir to prevent an accumulation of foreign material. Rectangular reservoirs shall be reinforced and supported in such a manner to insure that the top of the reservoir will not sag.

3.7 .2.1.3 Reservoir pressure. The reservoir shall withstand a minimum of 2 psi Internal pressure without leakage or permanent distortion.

3.7.2.1.4 Reservoir evacuation. When required to maintain pressures less than atmospheric, the reservoir shall withstand a pressure of 1/2 inch Hg absolute without permanent distortion or leakage.

3.7 .2.1.5 Reservoir drain lines. The reservoir drain line, at least 1 inch ID, for reservoirs up to 100 gallons capacity, 1 1/2 inches ID for reservoirs between 100 and 150 gallons and 2 inches ID for reservoirs over 150 gallons capacity shall be piped to the outside at the rear of the test unit. The drain line shall terminate with a standard pipe coupling flush with the rear of the stand, a minimum of 8 inches above the floor, and shall be sealed with a pipe plug for protection during shipment.

3.7 .2.1.6 Vents. Unless otherwise specified, all tank vents shall terminate at the top of the stand with a female pipe coupling provided for connection to the facility common venting system. vent lines shall be a minimum of 1 inch diameter. Vent lines shall be sealed with a pipe plug prior to shipment

* 3.7.2.1.7 Waste reservoir Unless otherwise specified a waste reservoir shall be provided. Where a waste reservoir is required, it shall be located within the inclosure and directly below the sump and above the outlet on the rear of the panel. It shall have a minimum capacity of 5 gallons. The outlet shall be located a minimum of 16 inches above the floor.

Source: https://assist.dla.mil -- Downloaded: 2026t0-20T10:09Z
Check the source to verify that this is the current version before use.

3.7.2.1.8 Porting. The return port shall be located as remotely as possible from the suction port and shall be designed in such a manner that the return flow is directed away form the suction port. The return fluid shall be introduced at least 1 inch below the minimum operating fluid level.

* 3.7 .2.1.9 Reservoir gages. Liquid level gages shall be provided on all reservoirs . Hydrostatic, mechanical, or electrical type gages shall be used. The gage shall indicate the liquid level at the 1/4, 1/2, 3/4 and full positions. The gage shall be properly marked to prevent overfilling of the reservoir and shall be accurate to within 5 percent. The gage shall be marked by a red line to indicate the minimum operating level of the fluid in the reservoir. The reservoir gage shall be located in an easily accessible location.

3.7.2.1.10 Manual fill connection. A fill pipe for manual filling of reservoir shall be located within the inclosure in an easily accessible location for manual filling. The pipe shall be a minimum of 1-1/2 inches ID and shall be equipped with an underwriter's approved type tank fill connection which includes a strainer, a sealed cap, and a scupper.

3.7.2.2 Filters. All filters shall be of a size and type as provided in the detail specification and be equipped with standard commercial type elements. The filter shall be capable of handling flows at least 100 percent in excess of the maximum rated flow capacity of the test stand without exceeding the specified allowable pressure drop.

3.7 .2.2.1 Location of filters. Filters shall not be installed in the boost pump inlet, nor submerged in the supply reservoir.

3.7.2.2.2 Air bleed. Filters shall be self purging, or provided with a l/8-inch valve connected to tubing to divert the purged fluid directly to the main reservoir.

3.7 .2.2.3 Differential pressure. All filters shall be equipped with a gage or gages to indicate the pressure drop across the filter and shall be located in an easily accessible position at the side or rear of the stand. Filters may be equipped with a warning device to indicate maximum allowable pressure drop through the filter when approved by the procuring agency.

3.7 .2.2.4 Strainer. A suitable strainer shall be provided between the reservoir and low pressure centrifugal boost pump inlet which shall be easily accessible for removal and cleaning.

3.7.2.3 Location of service connections. Service couplings shall be located in horizontal or diagonal relative positions with a minimum of 6 inches between each coupling to facilitate interconnection to the facilities services.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:09Z
Check the source to verify that this is the current version before use.

* 3,7.2,4 Sump drain. Drain pans and working areas which collect test fluid in small quantities during test or test set-up shall be piped to the outside of the test equipment unit. The drain line shall terminate with a pipe coupling located a minimum of 16 inches above the floor. Sump drains shall be a minimum of four feet long, 18 inches wide and not less than 26 inches nor more than 30 inches above the floor.

3.7.2.5 Line termination, All drain lines, fill lines (except the manual fill line), heating and cooling lines shall terminate flush with the outside and rear of the cabinet with standard female pipe couplings.

3.7.2.6 Plumbing. All lines, valves and fittings shall be of sufficient s ize to allow operation under maximum flow conditions without experiencing excessive pressure drops Throughout the system.

3.7 .2.6.1 Rigid lines. Rigid lines where required shall be of corrosion resistant steel or aluminum alloy, suitable for the application. Rigid lines, fittings and components shall withstand a test of 200 percent of maximum operating pressure without leakage or permanent distortion. The use of commercial black pipe and fittings for hydraulic flulds shall be prohibited. The use of rigid lines shall be limited to a minimum, unless otherwise specified. All pipe threads shall be of standard commercial quality NPT. Materials other than thread lubricant and polytet rafluoroethylene tape shall not be used for sealing threaded joints.

3.7 ,2.6.2 Rigid line mounting. Rigid lines shall not be attached to the panel mounted valves or fittings. when sizes and pressures permit the use of flexible hose.

* 3.7 .2.6.3 Fluid lines and fittings. Flexible hose and AN or MS type fittings. shall be used whenever possible. Flexible hose will not be used if it is proven that the dampening effects results in erratic critical pressure readings. The use of hose clamps as fluld line couplings shall be prohibited. All fluid lines, valves and fittings shall be a type rated by their manufacturer for the pressure and service to be encountered and shall, in addition withstand 200 percent of maximum operating pressure without leakage or permanent distortion

3.7,2,7 Fluid Velocity. The maximum fluid velocity shall not exceed 15 feet per second in operating valves and lines, Velocities in return and suction lines shall not exceed 5 feet per second. The maximum fluid velocity through any instrument shall be as required for proper operation.

3.7.2.8 Flexible hose. All flexible hose carrying test fluids shall be of the flexible steel braid type with AN or MS type coupling fittings installed on each end of the hose. All flexible hose shall be checked for electrical continuity. The maximum resistance of the steel braid and the hose couplings shall not exceed 100 ohms,

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:09Z
Check the source to verify that this is the current version before use.

3 .7.2.9 Auxiliary supports. All fittings, lines tubing and other components attached to the panel that, due to weight and size, might be damaged in shipment shall be provided with auxiliary supports or mounting brackets to minimize the possibility of damage to the panel and panel mount ings ,

3.7.2.10 Flange connections. Components such as valves and pumps may use flange connections with pipe thread outlets provided the pipe threads conform to the requirements of this specification.

3.7.2.11 Adapters. All valves, bulkheads, adapters, and other panel-mounted components shall be mounted by screws to tapped holes or to tapped inserts of standard commercial type.

3.7.2.12 Bulkhead ports. Unless otherwise specified, ports shall be AN (or commercial equivalent) female type, or standard female pipe thread (NPT).

3.7.2.13 Low pressure boost pumps. Low pressure centrifugal boost pumps shall be of a type using a carbon shaft seal rotating against steel, preferable with the steel surface stationary and the carbon seal rotating with the shaft. Pumps shall be of a design to readily permit easy replacement of seals.

3.7 .2.13.1 Shaft seals. Stuffing box type shaft seals shall not be used.

3.7.2.14 Mechanical refrigeration. When the desired test fluid temperature is less than 80 F, mechanical refrigeration equipment shall be supplied as a part of the test equipment.

3.7 .2.14.1 Refrigeration unit. The refrigeration unit, wherever possible, shall be a hermetically sealed (by fusion) type, and shall include a terminal box to the electrical connections.

3.7 .2.14.2 Capacity of unit. The refrigeration unit shall be of sufficient capacity to permit continuous operation of the test stand at maximum output, and shall not require more than 15 minutes to stabilize the test fluid at the desired temperatures under normal operating conditions.

* 3.7.2.15 Immersion heaters. Test fluid temperatures above ambient shall be obtained by the use of immersion heaters. Installation shall provide for the heater elements to be totally immersed in the fluid to a minimum depth of 6 inches when the fluid is at its lowest operating level. The well or reservoir in which the immersion heaters are installed shall be equipped with an NFPA approved Nationally recognized laboratory listed automatic over-temperature shut off switch that will automatically shut the heater off at 15°F above the maximum or reservoir operating temperature. The well or reservoir shall also be provided with a low fluid level switch that will shut down the test stand when the fluid level is less than 6 inches above the heater elements. The immersion heater shall be equipped with a manual temperature setting control (see 3.7.5.1.5).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:09Z
Check the source to verify that this is the current version before use.

* 3.7.2.16 Heat exchanger system. Cooling of test fluid shall be accomplished through standard commercial type heat exchangers using cold water where the desired test fluid temperature is above 80°F.

* 3. 7.2.16.1 Cooling. The heat exchangers shall be of sufficient size to provide the required cooling at all times when operating at an ambient temperature of 100 F using cooling water at 70 F with an operating pressure of 40 to 70 psig.

* 3 .7.2.16.2 Operating pressures. Each heat exchanger shall withstand a maximum operating pressure of 100 psig without leakage or permanent distortion. The pressure drop through the heat exchanger shall not exceed 10 psig at maximum flow for each fluid (cooling water and test fluid).

* 3. 7.2.16.3 Heat exchanger purging . Heat exchangers shall be equipped with purge systems to permit the release of air trapped in the cooling water and in the test fluid.

3.7,2.17 Warmup time. Initial warmup of the fluid from 60° to operating temperature shall not require in excess of 15 minutes.

* 3,7.2,18 Flow. Flow of cold water shall not exceed 1/2 GPM per horsepower input to hydraulic system,

3.7.2.19 Overpressure protection, Relied valves shall be provided on the discharge side of all pumps. These valves are required to prevent damage to the test stand or the item on test.

3.7.2.20 Accumulators . Accumulators shall be of the piston type and in general accordance with Specification MIL-A-5498 or a suitable industrial type . Means shall be provided for safety relieving accumulator gas and hydraulic pressures.

3.7.2.20.1 Accumulator pressure indication. A pressure gage shall be provided to indicate accumulator gas charge pressure.

3.7.2 .20.2 Recharging the accumulator. An AN6287 air valve or commercial equivalent connector shall be provided and installed in an easily accessible location for recharging the accumulator.

3 .7.2.20.3 Charge gas. In hydraulic systems where gas is used, nitrogen or other inert gas shall be used, unless otherwise specified.

3. 7.2.20.4 Surge tanks. Surge tanks which will permit the gas charge to come in physical contact with the hydraulic fluids shall not be used.

3.7.2 .21 Components. All components of the hydraulic system shall be suitable for the type of test fluid specified in the detail specification.

3.7 .2.22 Actuators. Direct air operated cylinders for actuation of any device shall not be used,

3.7.3 Pneumatic systems.

Source: https://assist.dla.mil -- Downloaded: 202640-20T10:09Z
Check the source to verify that this is the current version before use.

3 .7.3.1 Compressed air regulating system. A compressed air regulating system shall be provided, of adequate size and stability to perform the required tests.

3.7 .3.1.1 Air regulating units. There shall be provided a 0-200 psi gage to indicate pressure at source, an approved filter and trap, a 200 psi rated regulator, and a suitable gage to indicate regulated pressure.

3.7.3.2 High pressure air. Where high pressure air is required, the components to be furnished by the manufacturer shall be as specified in the detail specification.

3.7.4 Instruments.

3.7.4.1 Flowmeters. Flowmeters shall be as specified in the detail specification.

3.7.4.1.1 Scales. Meter tubes and scales of variable area glass tube type meters shall have logarithmic characteristics to provide uniform percentage readability at all points on the scale and shall be calibrated in gallons per minute with at least one half of one percent graduations for use with the type fluid specified in the detail specification. One half of one percent of instantaneous ‘flow at any point shall not be less than 0.0625 inch. Flow position readings to the nearest scale graduation shall not exceed one half of one percent of the indicated flow.

3.7 .4.1.1.1 Accuracy. The meter shall have an accuracy within one half of one percent when operating on a fluid within the specified density, viscosity and temperature limits.

3.7 .4.1.2 Line pressure. The meter shall be designed for a line pressure of 50 psi unless otherwise specified in the detail specification. The meter shall withstand a leak test of 150 percent of the maximum operating pressure.

3.7 .4.1.3 Pressure drop. Maximum pressure drop through the meter or through a series of meters shall not exceed 8 psi, When approved by the procuring agency, a back pressure valve may be used for suppression of vapor release and bubbles within the instrument.

3.7 .4.1.4 Vent valves. Unless otherwise specified the flowmeter shall have provisions for attaching vent valves to each tube which discharge through visible drip indicators to return to the main fluid reservoir. The vent valves, indicators, and return lines need not be component parts of the meter,

3.7 .4.1.5 Blowout plugs. Meters shall be protected by blowout plugs connected so that any leakage past the blowout plug will be indicated at a sight-f low indicator located on the front panel and in a line returning by passed fluid to the main reservoir. Blowout plugs shall be of the diaphragms type, using steel, brass, or synthetic diaphragms as outlined in the detail specificat ion. Plugs shall be installed in an easily accessible location and shall be designed for quick removal and installation of the blowout diaphragms.

Source: https://assist.dla.mil -- Downloaded: 202640-20T10:09Z
Check the source to verify that this is the current version before use.

3.7.4. 1.5.1 Blowout plug strainer. A strainer shall be provided in the down stream side of the blowout disc assembly to prevent the ruptured blowout plug from contaminating the system.

3. 7.4.1.6 Multiple tube meters, Multiple tube meters shall be assembled into a single unit including all necessary interconnecting pipes and valves. Meters shall be constructed to permit cleaning without removal of the tubes and designed so that the meters may be readily removed as a unit. The meter shall not require recalibration after cleaning and reassembly. An opening shall be provided at the top of the test stand to permit access for cleaning the meter tubes.

3.7.4. 1.7 Float guides, Floats utilized in flowmeter tubes shall be guided by the side wall of the tube, or through the use of a center guide wire. Construct Ion of float guides must be such as to permit removal of float and guide wire, if used, for cleaning without disassembly of the meter.

3.7 ,4.1.8 Calibration ports. Calibration ports shall be provided at the inlet and outlet of the flowmeter bank. The ports shall be provided with suitable bleeds and shall be at least five tube diameters distant from the inlet and outlet of the bank. Ports shall be readily accessible.

‘3 .7.4.1.9 Remote indicating flowmeters. Remote indicating flowmeters may be used in lieu of the variable area glass tube type. Accuracy shall be within one half of one percent of instantaneous reading, unless otherwise specified. Pressure drop shall not exceed 8 psi.

3.7.4.1.10 oil flowmeters. oil flowmeters shall be as specified in the detail specification. Unless otherwise specified, oil flowmeter shall be calibrated for readings in gallons per minute and shall have an accuracy of one percent of instantaneous reading. Pressure drop shall not exceed 8 psi.

3.7.4.2 Speed indication. When it is required that speed be read to an accuracy. greater than one percent of the indicated speed, a digital counter shall be used. A combination of tachometer, counter, and timer may be used when approved by the procuring agency.

3.7.4,3 Tachometer. The size, type, range and degree of accuracy of the tachometer or counter and timer shall be as outlined in the detail specification. Electrical components shall be easily removable by plug-in type unit construction and shall be contained in one chassis unit.

3 ,7.4.3.1 Calibrating Provisions shall be made for checking the calibration of tachometers other than electronic, without removal from the stand, either by checking the shaft speed with a hand tachometer, or by incorporation of a corm-non junction box with disconnect plugs for checking with auxiliary test equipment.

Source: https://assist.dla.mil -- Downloaded: 202610-20T10:09Z
Check the source to verify that this is the current version before use.

3.7.4.4 Timers. Where precision timers are required, the timer shall be self contained and shall not require an external source of frequency control,

3 .7.4.5 Temperature indicators. Temperature indicators shall be of a type and range as required by the detail specification. Dial type temperature indicators where used shall be mercury actuated with fully compensated capillary tubes.

* 3. 7.4.6 Pressure gages. All panel mounted gages shall be recess type provided with mounting flange at front of gage. The mounting flange (and dial) shall not extend more than 5/8 inch beyond the external surface of the panel. All gages shall incorporate a safety blowout device to prevent blowout through the face of the gage. The safety blowout device shall automatically relieve the gage case pressure should it reach 1/2 psi due to gage failure. The pointer shall be adjustable. Good quality stainless steel or nylon movements shall be required on all gages. Gages subjected to continuous surges and extreme pressure fluctuations shall be protected by a hydraulic dampener. The type, size, range and accuracy of pressure gages shall be as provided in the detail specification. The dial and pointer shall be separated from the gage movement by a wall which will prevent rupture through the front of the gage in the event of gage failure. The gage dial shall have a mirror surface to prevent parallax, and the pointer shall be turned to a 90 degree angle for accurate readability. Gages shall be provided with shatterproof glass.

3. 7.4.6.1 Mounting. Gages shall be mounted by screws and tapped panel holes to permit removal by one man from the front of the panel.

3. 7.4.6.2 Connection. Gages shall be connected to the panel with flexible AN or MS type hose or aommercial equivalent, of sufficient length to permit removal and installation from the front of the panel.

3.7 .4.6.3 Throttling. All pressure gages shall be equipped with needle valves, located as close as practicable to the gage.

* 3.7 .4.6.4 Protect Ion. Gages subject to pressure beyond their maximum range shall be protected with standard commercial type automatic gage shut-off valves. Automatic gage shut-off valves shall be of positive action which will allow absolutely no leakage of fluid to the gage. A suitable in-linee filter shall be provided to the circuit immediately ahead of the shut-off. The difference between the cut-off and cut-on point shall not exceed $\pm$2 percent of the range of the gage.

3.7.4.6.5 Air bleed. Where maximum accuracy is required in low pressure range gages and where the fluid head in the lines will cause a significant error in the gage reading, provision shall be made to free gage lines of air and maintain a constant liquid head on the gage.

3. 7.4.6.6 Drip indictor. Bleed lines on pressure gages shall be provided with visible drip indicators.

Source: https://assist.dla.mil - Downloaded: 2026l0-20T10:09Z
Check the source to verify that this is the current version before use.

3.7.4.6.7 spuds. All pressure gages shall be provided with a tap on the panel connected between the gage and the gage shut-off valve to permit calibration of the gage without removal from the panel.

3.7,4.7 Manometers. Absolute pressure manometers used for setting up density shall be of the density calculating type and shall be calibrated for the densities specified in the detail specification

3.7.4.7. J Material. All metal parts of manometers in contact with the indicating fluid used shall be of stainless steel.

3.7.4.7.2 Pressure. All manometers shall withstand 125 percent of normal operating pressure imposed without leakage or permanent distortion .

3.7.4.7.3 Protectors. Suitable valves (which include a fluid trap) that will prevent the indicating fluid from being forced out of the manometer shall be installed at the top of the manometer column and in the wall connection.

3.7.4.7,4 Type. Type and range of manometer shall be as provided in the detail specification.

3.7,4.8 Torque measuring equipment. Where it is necessary to measure the torque required to drive the item on test, a direct indicating torquemeter shall be provided. The meter shall read in measurement values as outlined in the detail specification.

3,7.5 Controls

3.7.5.1 Temperature control. Automatic controls shall be provided for heating and cooling the test fluid and shall maintain the set temperature within plus or minus 2°F. The type of automatic control shall maintain the temperature within the required limits during all conditions normally encountered during test.

3,7.5.1,1 Temperature range. The range throughout which the automatic temperature control can be adjusted shall be as stated in the detail specification.

* 3.7.5,1.2 Operation. The temperature of the cooling medium shall be as specified in 3.7.2.16.1; however, the automatic temperature control must also operate successfully with cooling water of a lower temperature then specified or with the manual temperature setting control of the immersion heater set at a higher temperature than necessary

3.7 .5.1,3 Response. The response of the automatic temperature control shall be fast enough to maintain the temperature within the required limits during transient conditions to permit uninterrupted testing.

3.7 ,5.1,4 Acceptance. At the option of the procuring agency, a thermocouple and potentiometer may be used to measure the temperature during acceptance tests of the temperature control system.

Source: https://assist.dla.mil - Downloaded: 2026-10-20T10:09Z
Check the source to verify that this is the current version before use.

3.7.5.1.5 Manual Temperature control. Manual controls shall be furnished only to trim the cold water supply for better flow control and for use as shut-off valves, and to set the maximum temperature output of the immersion heaters,

3.7.5.2 Manual operating controls. All controls shall be selected and arranged to provide a maximum of convenience and operating ease, to reduce test time and operator fatigue with the ultimate prupose of obtaining maximum production from the test equipment. Controls may be located on the horizontal areas adjacent to the sump drain, when approved by the procuring agency .

3.7.5.3 Location. All controls, instrumentation, and accessory drives shall be located so as to permit all phases of the test to be accomplished by one operator from the front of the test unit. Those instruments which affect the accuracy or results of the test shall be located for ease of observation and readability.

3.7.5.4 Remote operating, Unless otherwise specified in the detail specification, control valves in excess of 3/4 inch port size, which due to operating pressure and frequency of use will cause extreme operator fatigue, shall be remotely operated by air with a regulat ing valve located on the front panel. The operator section of the control valve shall be capable of withstanding continuous service at 150 psi air. The control shall be such that in the event of current or air failure the valve will be returned to “rest” or “safe” position.

3.7.5.5 Automatic controls. Where it is necessary to control pressure of flow to a constant value, automatic controls shall be used. A slight amount of adjustment of the control will be permitted when it is necessary to control within close linits and accuracy as outlined in the detail specification.

3.7.6 Valves

3.7.6.1 Valves. All valves shall be of corrosion resisting material unless otherwise specified.

3.7.6.2 Relied valves. High pressure relief valves shall be of the pilot operated type of construction. The pressure setting shall not drop below 20 percent of setting. Differential of relief valves shall not exceed 15 percent of setting. Relief valves shall not chatter,

3.7 .6.2.1 Metered lines. Relief valves installed in metered lines shall indicate any leakage through a visible drip indicator located on the front panel.

* 3.7.6.3 Shut-off and control valves. The valves shall be standard industrial catalog items and of corrosion resistant materials compatible to the fluids specified. Handle loads shall be minimized by the use of ball thrust bearings or by balanced pressures. Detents or nameplates shall be provided to identify correct handle positions. The flow passage diameters in the valves shall be equal to or greater than the internal diameter of the piping to which they are connected. The valves shall so operate that no undue hydraulic shocks or surges shall result from their use to change flow volume or direction. The maximum allowable torque required to operate manual control valves shall not exceed 40 pound-inches. Push-pull types of valves

Source: https://assist.dla.mil -- Downloaded: 202610-20T10:09Z
Check the source to verify that this is the current version before use.

may be used provided they meet the applicable requirements.

3.7.6.4 Control valves, Control or throttling valves shall be of linear flow type suitable for the intended application.

3.7.6.4.1 Hysteresis. No hysteresis shall be present and valves shall not creep from selected position.

3,7.6.5 Needle valves. Needle valves shall be panel mounted type with stainless steel stems. All needle valves used shall show the manufacturers' cold working pressure rat ing stamped on the valve body as well as arrow shining flow direction. NO commercial needle valve shall be applied to pressure beyond the manufacturers' rating Needle and packing shall be readily accessible from the front of the stand. A valve knob or handle shall be installed on a square or section flat section of the valve stem. Valves shall be of a type that can be disassembled for inspection repair or replacement of parts.

3.7.6.6 Automatic shut-off valves. Shut-off valves shall be interlocked to operate automat ically with the test stand emergency disconnect controls. Shut-off valves other than those in the flow-meter system shall be standard commercial tyoe air operated or solenoid actuated valves. In the event of a hydraulic unit, power, water or air failure they shall fall in the safe position.

3.8 Vacuum system Where vacuum is necessary to meet test requirements, unless otherwise specified, a complete self–contained vacuum system shall be furnished by the equipment manufacturer. This system shall consist of an electric motor and vacuum pump unit of sufficient capacity and stability to maintain the test conditions.

3.9 Lubrication. All components requiring periodic lubrication shall be provided with standard commercial type fittings suitable for the required lubricant.

3.9.1 Location of fittings. Fittings shall be installed in easily accessible locations and a placard shall be provided adjacent to each fitting indicating the type of lubricant to be used. Components installed within the inclosure at Inaccessible locations shall be connected by commercial tubing (rigid or flexible) to an adjacent location outside the structure and above the level of the bearing to be lubricated to permit flow of the lubricant by gravity.

3.10 Schematics. Electrical, lubrication, and hydraulic, schematics of the test unit shall be furnished in plastic sealed envelopes or equivalent and shall be permanently attached to the test stand. Schematics shall be clear and easily interpreted by using standard codes, and shall be plainly visible through the envelope. Schematics shall be a minimum of 8 inches by 10 inches. Test stands built in sections shall have sectional schematics in addition to the composite schematic.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:09Z
Check the source to verify that this is the current version before use.

3.11 Accessories. Parts such as gaskets, seals, "O"-rings and blowout plugs which will normally be required during the initial thirty days of operat ion (Including filter cartridges) shall be provided with the test unit

3.12 Interchangeability. All parts having the same manufacturer’s part number shall be functionally and dimensionally interchangeable,

* 3.13 Finish and protective coatings. Cleaning, painting, plating, and anodic films shall be in accordance with MIL-STD-808.

* 3.13.1 Exterior color. Exterior surface of the test stands, except where stainless steel is specified, shall be semi-glnss gray number 26373 conforming to FED-STD-595.

* 3.13.2 Interior color. Interior of the test stand shall be semi-gloss white number 27886 conforming to FED-STD-595.

* 3.13.3 Cadmium plating. Cadmium plating shall be prohibited on components of the hydraulic and pneumatic test circuits and on test stand components that can come into contact with aircraft items to be tested,

3.14 Identification of Product.

3.14.1 Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

* 3.14.2 Nameplates. Nameplates shall be attached by metal screw or rivets. The nameplate shall include at least the following information:

a. Procuring Agency Nomenclature
b. procuring Agency Stock Number
c. Contract Number
d. Vendor’s Name
e. Vendor ‘ Part Number
f. Vendor's Serial Number
g. Manufacturer’s Part Number
h. Property U. S. Government

3. 14.2.1 Vendor’s nameplate. The vendor’s standard nameplate may also be installed in addition to the procuring agency identification plate.

3.14.3 Placards. All valves, gages, outlets, connections and controls shall be identified by a name and number placard on the panel. The same numbers shall be shown on the flow diagram and other data furnished with test unit.

3. 14.3.1 Warning. Warning placards or other precautionary labels as necessary shall be so Located and designed that the operator cannot fail to notice them before any controls, valves or switches are operated. Warning placards shall be of laminated phenolic sheet with red surface backing to result in a red placard with white engraved letters.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:09Z
Check the source to verify that this is the current version before use.

3.14.4 Service lines. Steam, water, air, and drain lines shall be marked at entrance to the equipment, and shall be located at the rear of the test equipment unit.

3.14.5 Mounting adapters. All mounting adapters and drive couplings shall be identified by a separate manufacturers' part number stamped on each item.

3.14.6 Rotation. Directions of rotation on all components shall be properly identified with engraved or painted arrows.

3.15 Workmanship

3.15.1 The test stand, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner in accordance with high grade manufacturing practice for this type of equipment.

3.15.2 Neatness. Particular attention shall be given to neatness of wiring, installation of tubing, hose, and fittings; thoroughness of welding, brazing, drilling, forming, riveting and painting; alignment of parts and tightness of screws and bolts; and freedom from defects, burrs, and sharp edges.

* 3.15.3 Cleaning. The vendor shall be required to completely clean the test stand , and flush the entire hydraulic system with Specification MIL-H-5606 fluid, prior to preproduction inspection and preproduction tests.

3,16 Modified items. Items modified for a new purpuse or a higher rating than recommended by their manufacturer shall require approval of the procuring activity before being incorporated into the test unit.

† 3.17 Reclaimed materials. The use of reclaimed materials shall be encouraged to the maximum extent possible.

4. QUALITY ASSURANCE PROVISIONS

4,1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein, Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements,

4.1.1 Classification of tests. The methods of sampling, inspection and tests conducted upon test equipment units shall be classified as follows:

a. Acceptance Tests
b, Preproduction Tests

4,2 Acceptance tests

4.2.1 The acceptance tests shall consist of the following individual tests as described under 4.4 Test Methods of this specification:

a. Examination of Product
b. Performance Test
c. Correlation Tests

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:09Z
Check the source to verify that this is the current version before use.

4.3 Preproduction tests

4.3.1 preproduction tests shall be made on a sample(s) representative of the production test equipment units. Test shall consist of all tests described under 4.4 Test Methods.

4.4 Test Methods

4.4.1 Examination of product. Examination of product shall be made on production test units submitted for acceptance under the contract to determine conformance with the requirements of this specification and the detail speci fication.

4.4.2 Performance tests. All test equipment units shall be operated at full rated capacity for a minimum of 15 minutes and shall maintain proper test fluid temperature, speed, and speed control with no evidence of overloading or other indication of undue stress or failure. The test conditions being described in the detail specification.

4.4.3 Correlation tests. When desirable, the detail specification shall describe a test suitable for correlation and performance acceptance for the production quantities. Unless otherwise specified, correlation test shall be run on master or standard units previously calibrated by the prime contractor or by an organization designated and approved by the procuring agency. Corre- lation shall be within acceptable limits as outlined in the detail specifica- tion.

* 4,5 Examination of packaging . Examination shall be made to determine that packaging, packing and markings comply with the requirements of Section 5.

5. PACKAGING

* 5.1 Preservation and packaging. Preservation and packaging shall be level A or C in accordance with MIL-STD-794 as specified (see 6.2).

5.1.1 Level A. Test stand(s) shall be perserved and packaged one each in accordance with Specification MIL-P-116, Method IA-16.

5.1.2 Level C. unless otherwise specified the test stand(s) shall be preserved in a manner which will insure arrival at destination in serviceable condition.

5.2 Packing. Packing shall be level A, B or C in accordance with MIL-STD-794 as specified (see 6.2).

5.2.1 Level A. Test stands preserved and packaged as specified in para- graph 5.1.1 shall be packed in accordance with Specification MIL-STD-794. Cushioning, blocking and bracing shall be in accordance with MIL-STD-1186 Exterior container(s) shall be export type in accordance with ‘pacification MIL-C-104 Type II. Class and style optional. As far as practical, the crates shall be of uniform size and minimum weight and cube,

5.2.2 Level B. Same as 5.2.1 above, except exterior containers shall be of the domestic type,

Source: https://assist.dla.mil Downloaded: 2026-10-20T10:09Z
Check the source to verify that this is the current version before use.

5.2.3 Level C. Test stand(s) shall be packed in a manner which will insure arrival at destination in serviceable condition and be acceptable to the carrier at lowest rates. Containers and packing shall comply with Uniform Freight Classification rules or National Motor Freight Classification rules.

5,3 Marking. In addition to any special markings required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. Notes

6.1 Intended use. The test stand covered by this specification is intended for use in ground checking and maintenance of aircraft hydraulic components.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number , and date of this specification
b. Applicable level of preservation and packaging and packing

6.2.1 Preproduction. Samples will be required to be subjected to the preproduction tests to determine compliance with the requirements of the specification.

6.2.2 Installation data. In addition to any other data requirements in the detail specification or covered by a separate contract, one copy of commercial installation data, including operating and maintenance instructions and parts breakdown shall be packed with each test equipment unit. Three additional copies for each unit (or a minimum of 25 copies, whichever is the greater) shall be shipped to the procuring agency in accordance with shipping instructions to be furnished to the test equipment manufacturer.

6.2.3 Commercial literature One complete set of commercial literature or handbooks covering all the commercial components of the basic test stand, normally furnished by the manufacturer of commercial vendors' items, shall be packed with each test unit in looseleaf form. The handbooks or commercial literature shall contain the manufacturer's operating and service instructions. one complete set shall also be furnished the procuring activity with each test unit .

6.3 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Air Force - 99

Preparing Activity:
Air Force - 82

Reviev Activities:
Army - AV
Navy - AS

Project No. 4920-F370

Source: https://assist.dla.mil - Downloaded: 2026-10-20T10:09Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS This form is provided to solicit beneficial comments which may improve this document and enhance its use DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government Fold on lines on reverse side, staple in corner, and send to preparing activity Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed

NOTE This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements

DOCUMENT IDENTIFIER *(Number)* AND TITLE
MIL-S-24776A

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1 ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDED WORDING CHANGE

C REASON FOR RECOMMENDED CHANGE(S)

2 REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO

DATE

DD FORM 1 OCT 76 1426 PREVIOUS EDITION WILL BE USED

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:09Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE AIR FORCE

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

BUSINESS REPLY MAIL

FIRST CLASS PERMIT NO 73236 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE AIR FORCE

SA-ALC/MMEDO
Kelly AFB, TX 78241

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T10:09Z
Check the source to verify that this is the current version before use.